HAZARD IDENTIFICATION, RISK ASSESSMENT, RISK MANAGEMENT AND CONTROL MEASURES

THE Hazard: The "Source" of Harm

- A hazard is an inherent property of an object, substance, or situation. It exists regardless of whether anyone is nearby
- A hazard is often "waiting" to happen.

e.g., a frayed wire

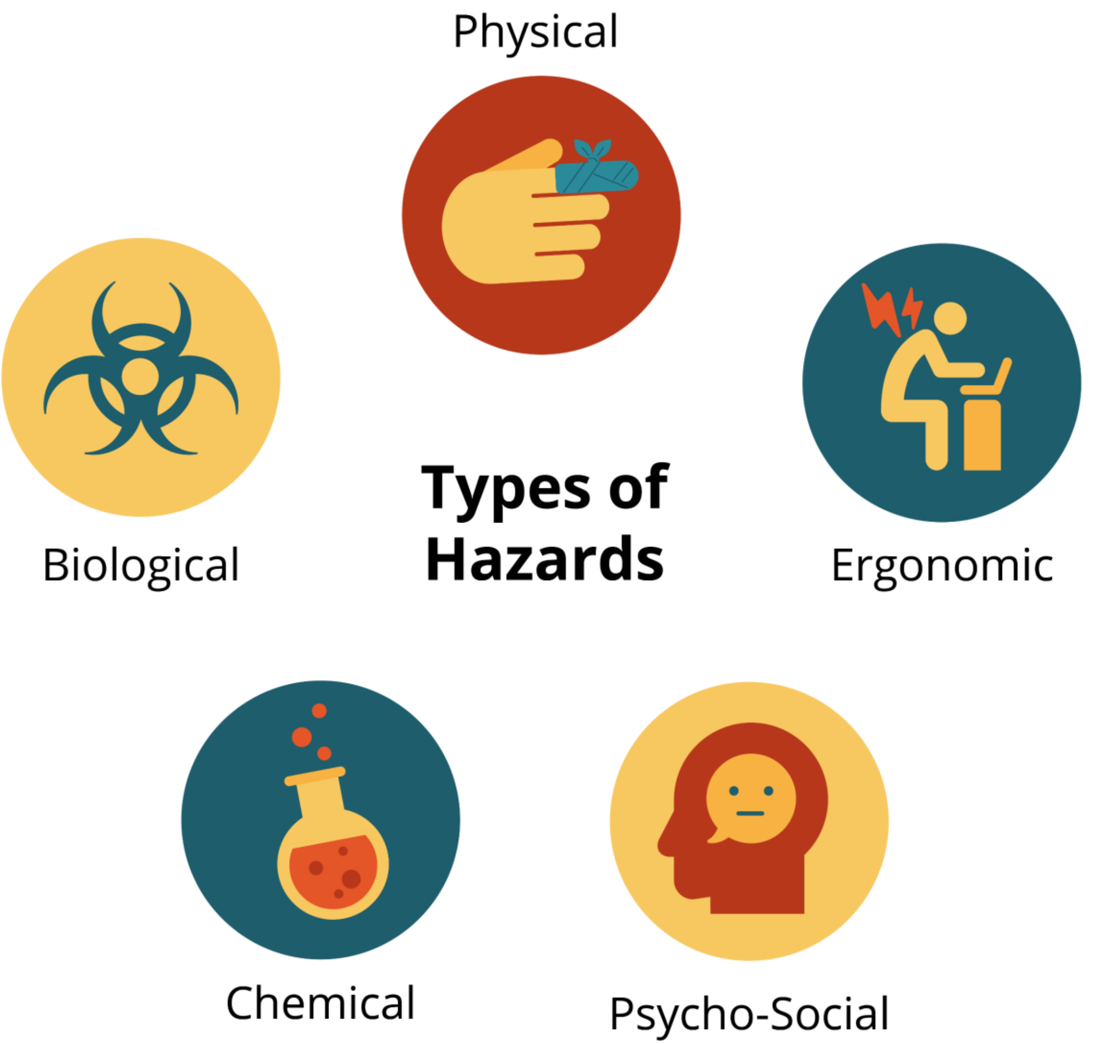
THE RISK: THE "PROBABILITY" OF HARM

- The *likelihood* that the harm will occur, combined with its severity

e.g., how likely is someone to touch that wire, and will it tingle or kill them?


HAZARD
vs
RISK
A HAZARD is something that has the potential to harm you
RISK is the likelihood of a hazard causing harm


Hazard
Vs
Risk
A hazard is anything that can cause harm.
A risk is the likelihood of harm from a hazard.
Certainty

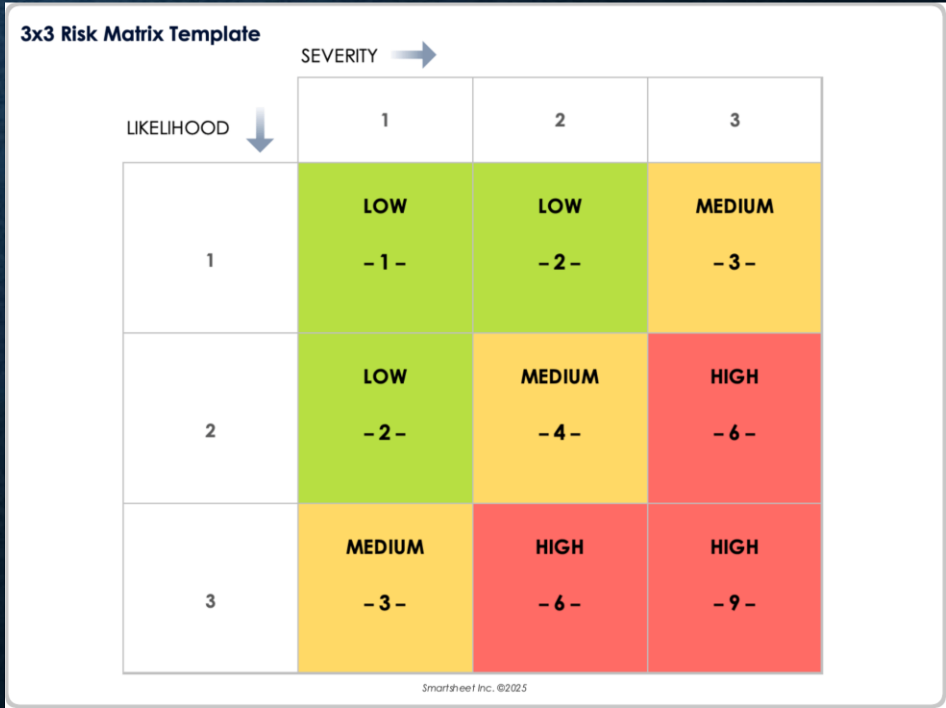

Physical
Biological
Types of Hazards
Ergonomic
Chemical
Psycho-Social

TYPES OF HAZARDS

Physical: Noise, radiation, extreme temperatures, unguarded machinery, heights.
Chemical: Corrosives, toxins, gases, cleaning fluids, vapors, battery acid.
Biological: Pathogens, viruses, bacteria, mold, bodily fluids.
Ergonomic: Heavy lifting, repetitive twisting, poor workstation setup
Psychosocial: Workplace bullying, extreme fatigue, Stress, workplace violence, or harassment.

RISK CALCULATION

- It is dynamic and changes based on the environment and human behavior. It is expressed by the formula:
- Risk = Probability (Likelihood) × Severity (Consequences)



3x3 Risk Matrix Template

LIKELIHOOD ↓ / SEVERITY →	1	2	3
1	LOW – 1 –	LOW – 2 –	MEDIUM – 3 –
2	LOW – 2 –	MEDIUM – 4 –	HIGH – 6 –
3	MEDIUM – 3 –	HIGH – 6 –	HIGH – 9 –

LIKELIHOOD (PROBABILITY)

3x3 Risk Matrix Template

LIKELIHOOD ↓ \ SEVERITY →	1	2	3
1	LOW - 1 -	LOW - 2 -	MEDIUM - 3 -
2	LOW - 2 -	MEDIUM - 4 -	HIGH - 6 -
3	MEDIUM - 3 -	HIGH - 6 -	HIGH - 9 -

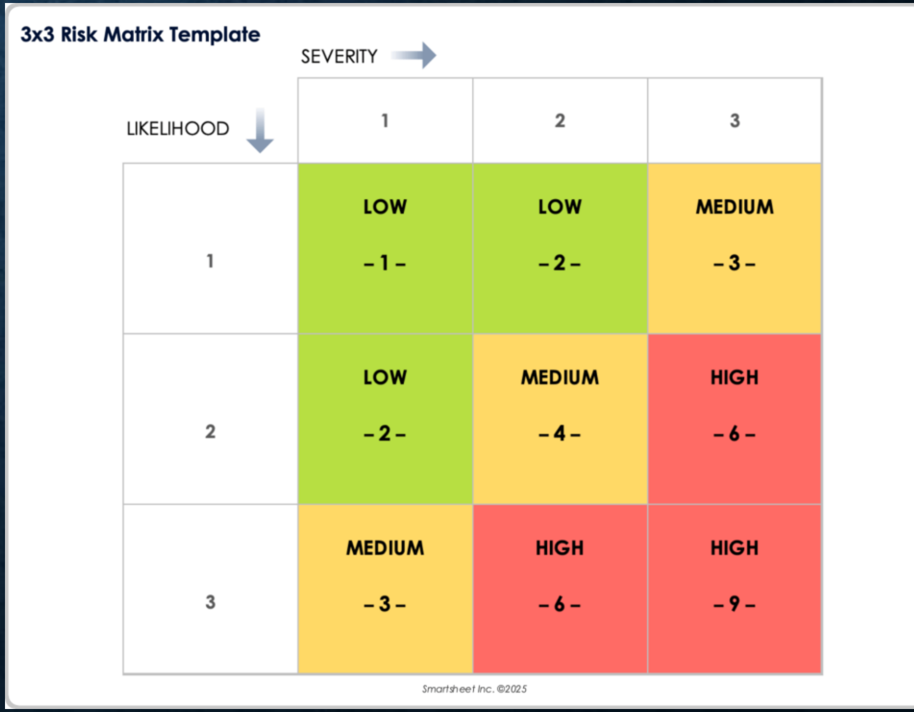
- How often is the task performed?
- How many people are exposed?
- What is the duration?

- *Low:* A frayed wire tucked behind a heavy server rack that no one ever moves.
- *High:* A frayed wire on a handheld vacuum cleaner used daily by the cleaning staff.

SEVERITY (CONSEQUENCE)

- If the event happens, how bad is it?

- *Minor:* A small shock or "tingle."“
- *Major:* Cardiac arrest, fire, or permanent nerve damage.



3x3 Risk Matrix Template

SEVERITY →

LIKELIHOOD ↓	1	2	3
1	LOW – 1 –	LOW – 2 –	MEDIUM – 3 –
2	LOW – 2 –	MEDIUM – 4 –	HIGH – 6 –
3	MEDIUM – 3 –	HIGH – 6 –	HIGH – 9 –

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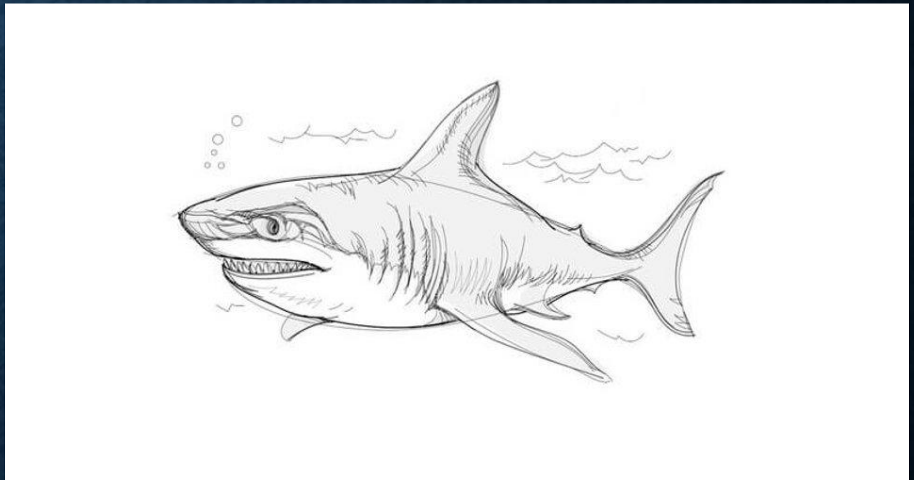
SHARK IN THE WATER

- **The Hazard:**

A Great White Shark. It is inherently dangerous; it has teeth and it hunts.

The Risk:

- If the shark is in the ocean and you are on the beach, the **Risk is Low** (low likelihood of contact).
- If you jump into the water to swim with the shark, the **Risk is High** (high likelihood of contact + high severity of injury).



ANALYZE THE FOLLOWING SCENARIOS TO IDENTIFY THE CORE HAZARD AND DETERMINE IF THE RISK LEVEL IS LOW OR HIGH BASED ON THE LIKELIHOOD AND POTENTIAL SEVERITY OF AN ACCIDENT.

SCENARIO A: FOOD PREPARATION

- An untrained worker is using a dull knife to cut a frozen object while holding the item in their hand.

- **The Hazard:** A dull blade and improper cutting technique
- **Risk Level:** High
- **Reasoning:** A dull blade requires more force, increasing the chance of it slipping. Since the object is being held in the hand rather than on a stable surface, the likelihood of a severe laceration is extremely high

ANALYZE THE FOLLOWING SCENARIOS TO IDENTIFY THE CORE HAZARD AND DETERMINE IF THE RISK LEVEL IS LOW OR HIGH BASED ON THE LIKELIHOOD AND POTENTIAL SEVERITY OF AN ACCIDENT.

SCENARIO B: FACILITY MAINTENANCE

- A liquid spill located inside a locked janitor's closet with no foot traffic.

- **The Hazard:** A slippery surface / Spil
- **Risk Level:** Low
- **Reasoning:** While the hazard exists, the likelihood of someone slipping is nearly zero because the area is locked and inaccessible to the general public or staff.

ANALYZE THE FOLLOWING SCENARIOS TO IDENTIFY THE CORE HAZARD AND DETERMINE IF THE RISK LEVEL IS LOW OR HIGH BASED ON THE LIKELIHOOD AND POTENTIAL SEVERITY OF AN ACCIDENT.

SCENARIO C: WORKING AT HEIGHT

- A worker is standing on the top rung of a 3-meter A-frame ladder placed on uneven ground.

- **The Hazard:** Working at height on an unstable base
- **Risk Level:** High
- **Reasoning:** Standing on the top rung shifts the center of gravity dangerously high, and the uneven ground makes a tip-over highly probable. A fall from 3 meters can result in "Major" severity, such as broken bones or head injuries.

The Hazard	Low Risk	High Risk
A Knife	A professional chef using a sharpened knife on a stable cutting board.	An untrained worker using a dull knife to cut a frozen object while holding it in their hand.
Wet Floor	A spill in a locked janitor's closet with no foot traffic.	A spill in the main entrance of a hospital during a shift change.
Height	A worker on a 3-meter scaffold with full guardrails and a harness.	A worker standing on the top rung of a 3-meter A-frame ladder on uneven ground.

Hazard stays the same, but the **Risk** changes based on context.

REPORT WORKPLACE HAZARDS

1. **The Immediate Response**
   - **Warn Others:** Verbally alert coworkers in the vicinity (e.g., "Watch out, there is oil on the floor!").
   - **Contain the Hazard:** If you can safely block off the area (using a cone or a chair) or clean a small spill, do it immediately.
   - **Emergency Stop:** If the hazard is life-threatening (e.g., a gas leak or exposed high-voltage wire), trigger the emergency alarm or shut down the machine immediately.
2. **Who to Report To?**
   - **Immediate Supervisor/Manager**
   - **Safety Officer/WHS Representative**
   - **Health and Safety Committee**

REPORT WORKPLACE HAZARDS

- **3. The "3-W" Reporting Method**
  - **What:** Describe the hazard specifically
  - **Where:** Be exact about the location.
  - **When:** The date and time the hazard was spotted.

GROUP WORK: WORKPLACE HAZARD ANALYSIS

Each group must analyze their assigned scenario. You must

- identify the hazard, and type
- determine who is affected,
- calculate the risk using the 3x3 matrix provided,
- **propose solutions using the Hierarchy of Controls.**
- **And monitoring and evaluation**

GROUP 1: THE COMMERCIAL KITCHEN (HOSPITALITY)

- During the lunch rush, a deep fryer's oil filtration system begins to leak, leaving a large, shimmering pool of hot oil on the tile floor directly in front of the main pass-through where servers collect plates. The kitchen is understaffed, and the Head Chef tells everyone to "just walk around it" until the rush is over.

GROUP 2: THE CONSTRUCTION SITE (INFRASTRUCTURE)

- A worker is operating a jackhammer on a highway expansion project. The temperature is (38c), and the worker has been at it for 4 hours without a scheduled break. There is no shaded area provided, and the worker has forgotten their water bottle in their personal vehicle parked

GROUP 3: THE GENERAL HOSPITAL (HEALTHCARE)

- In a busy emergency ward, several "sharps" containers (for used needles) are overflowing. Because the waste disposal contractor is late, a nurse attempts to push the needles down with a plastic ruler to make more room for a new syringe.

GROUP 4: THE TECH STARTUP (CORPORATE OFFICE)

- **Scenario:** To create a "cool" aesthetic, a new office has installed glass walls without any decals or frosted markings. In the first week, two employees have already walked into the glass. Additionally, many employees are using "standing desks" that have various power cables trailing across the floor to reach the nearest wall outlet.